

Multiple Choice (10 marks)

1. What is the output of "abc"[:-1] in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c
2. Let x = 1. What is x << 3 in Python 3?
 - (a) 1
 - (b) 3
 - (c) 8
 - (d) 16
3. Let x = 8. What is x >> 1 in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8
4. In Python 3, which of the following function convert a string to an int in python?
 - (a) int(x [,base])
 - (b) long(x [,base])
 - (c) float(x)
 - (d) str(x)
5. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.

True / False (10 marks)

Answer True or False

6. True or False: $O(n^2)$ is the largest asymptotic notation among $O(1)$, $O(n)$, $O(n^2)$, and $O(\log n)$.
7. True or False: In Python, the output of `print(list[1:3])` for the list `[ 'abcd', 786 , 2.23, 'john', 70.2 ]` is `['abcd', 786, 2.23, 'john', 70.2]`.
8. True or False: In Python 3, using `b[::2]` on the list `b = [11,13,15,17,19,21]` will output `[11,15,19]`.
9. True or False: The user's public key used for encryption poses the greatest risk to personal privacy among the listed options.
10. True or False: In Python 3, the lambda function `r` outputs 2 when called with the argument 3, as it divides the input value by 1.5.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a large Java program passing extensive testing without errors. What does this suggest about potential bugs, and why is it important to consider this possibility?
12. Discuss the purpose and functionality of the `'isupper()'` function in Python 3. How does it determine whether all characters in a string are uppercase, and why is this useful in programming?

End of Paper